

Isochrone & accessibility modeling

What this tool produces · *Geographical "catchment" maps (isochrones) that visually demonstrate the exact area reachable from a specific point within a given time limit (e.g., 10 or 15 minutes) via walking, cycling, or transit.*

G1 · Proof of Impact

Dimensions

Spatial

Social / User

Preparing the workshop · *Isochrone & accessibility modeling*

G1 · Proof of Impact

Why this tool matters

Demanding the right to time; using harsh geographical evidence to demonstrate that forcing residents to lose hours of their lives in transit is a spatial injustice that planning must correct.

Evidence we produce

Geographical "catchment" maps (isochrones) that visually demonstrate the exact area reachable from a specific point within a given time limit (e.g., 10 or 15 minutes) via walking, cycling, or transit.

Duration & Material

Duration · 2 to 4 weeks of geospatial analysis

Material · Routing engines (Valhalla, QGIS), GTFS public transit data, OpenStreetMap networks, census data

Reference case

To evaluate the 15-minute city framework in Budapest, Helsinki, and Madrid, researchers utilized the Valhalla routing engine to compute isochrones based on OpenStreetMap walking networks and GTFS transit schedules. By overlaying socio-economic data on these travel-time catchments, the study provided concrete evidence on how public transport expansions can support socio-economic integration in peripheral neighborhoods.

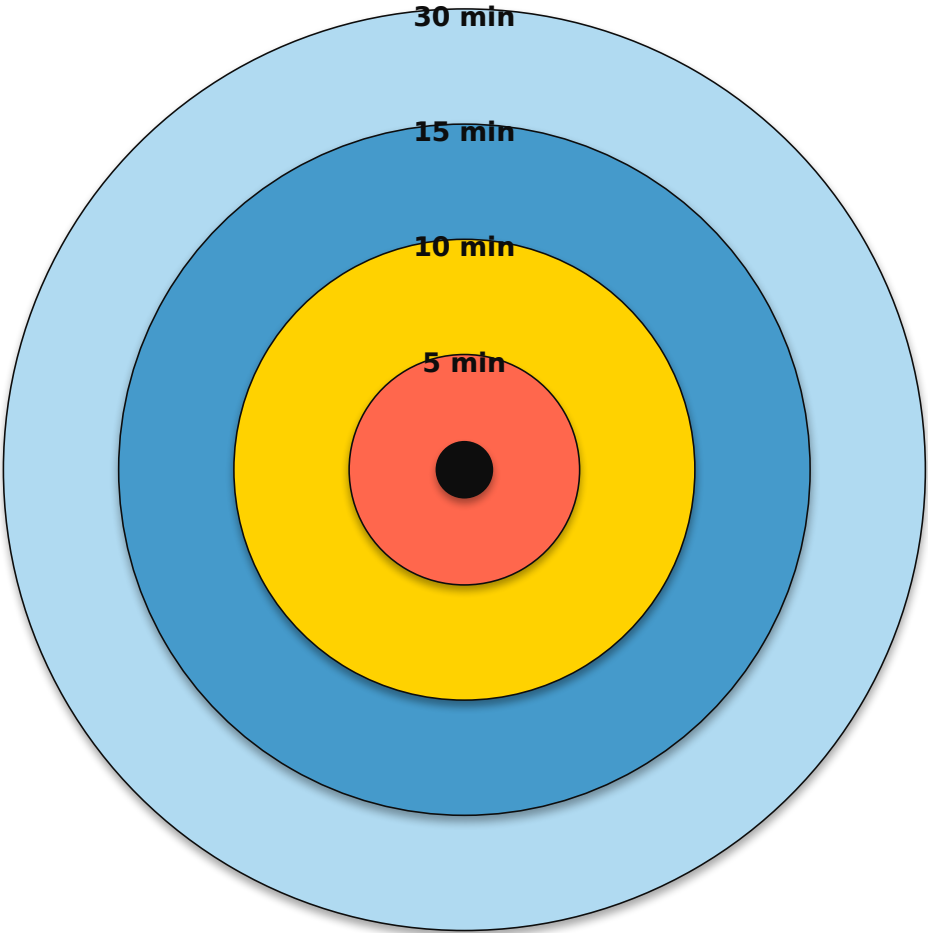
Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Isochrones · 5 / 10 / 15 / 30 minutes

Duration
2 to 4 weeks of geospatial analysis

Reach by mode

-  **Walking**
-  **Cycling**
-  **Public transport**
-  **Driving**



Accessibility notes

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.